Sales Team and Inventory Dashboard

Power BI project | 2-page dashboard | Sales of 2025

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Objective

Build a dashboard for a home-appliance company that tracks sales reps against their targets and monitors stock levels, so managers can coach reps and re-order products on time.

Business questions

• Which reps are above or below their annual target?
• Which regions, quarters and products sell best?
• Which products are running low on stock and how fast do they sell?
• Are there products nobody buys?

Key findings

• Total sales were Rs 26.71M from 520 deals, which is 98.0% of the combined annual target.

• 5 of 12 reps exceeded their target; 4 were near target and 3 were below. Best: Ayesha Butt (123%); lowest: Hamza Sheikh (88%).

• Q3 was the strongest (Rs 9.6M) and Q1 the weakest (Rs 5.4M).

• North is the top region (Rs 8.0M) and West the lowest (Rs 5.1M).

• Air Conditioner earns the most revenue (Rs 6.7M).

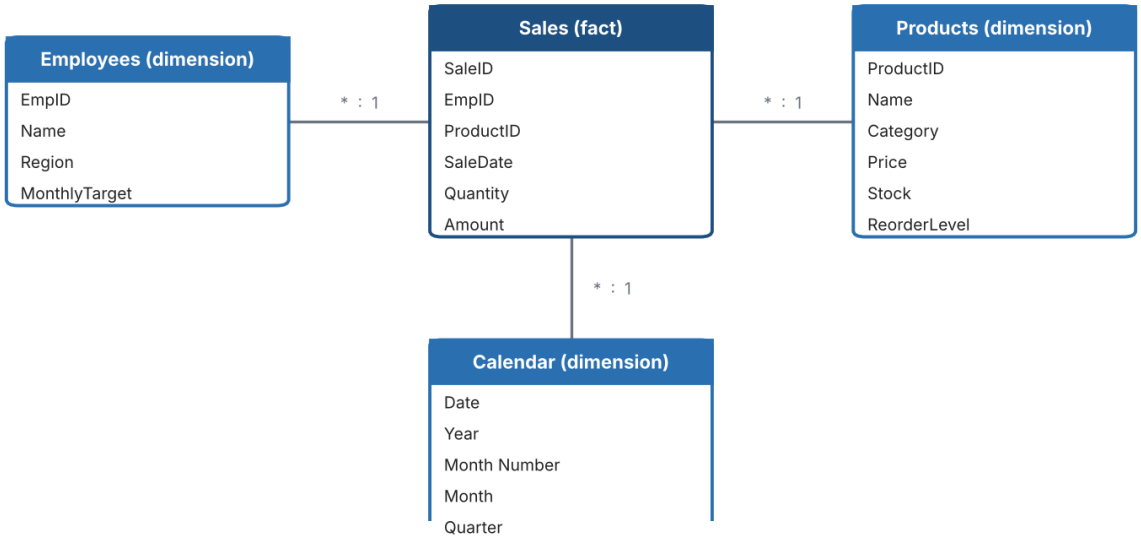
• 6 products are at or below their reorder level; the most urgent is Electric Kettle (stock 11, 45 units sold in the last 90 days).

• Steam Cleaner had no sales in the year.

Note: The dashboard previews in this report are drawn from the same sample data with Python to show the intended design and the correct numbers. Power BI's own visuals will look similar but not identical. The data is sample data created for this project; all amounts are in Rs.

1. Data model

Star schema: one fact table (Sales) with Employees and Products as dimensions plus a Calendar date table.



Relationships

Many side	One side	Direction
Sales[EmpID]	Employees[EmpID]	Single (one-to-many)
Sales[ProductID]	Products[ProductID]	Single (one-to-many)
Sales[SaleDate]	Calendar[Date]	Single (one-to-many)

After creating the Calendar table, right-click it and choose **Mark as date table**. Sort the Month column by Month Number so months appear in calendar order.

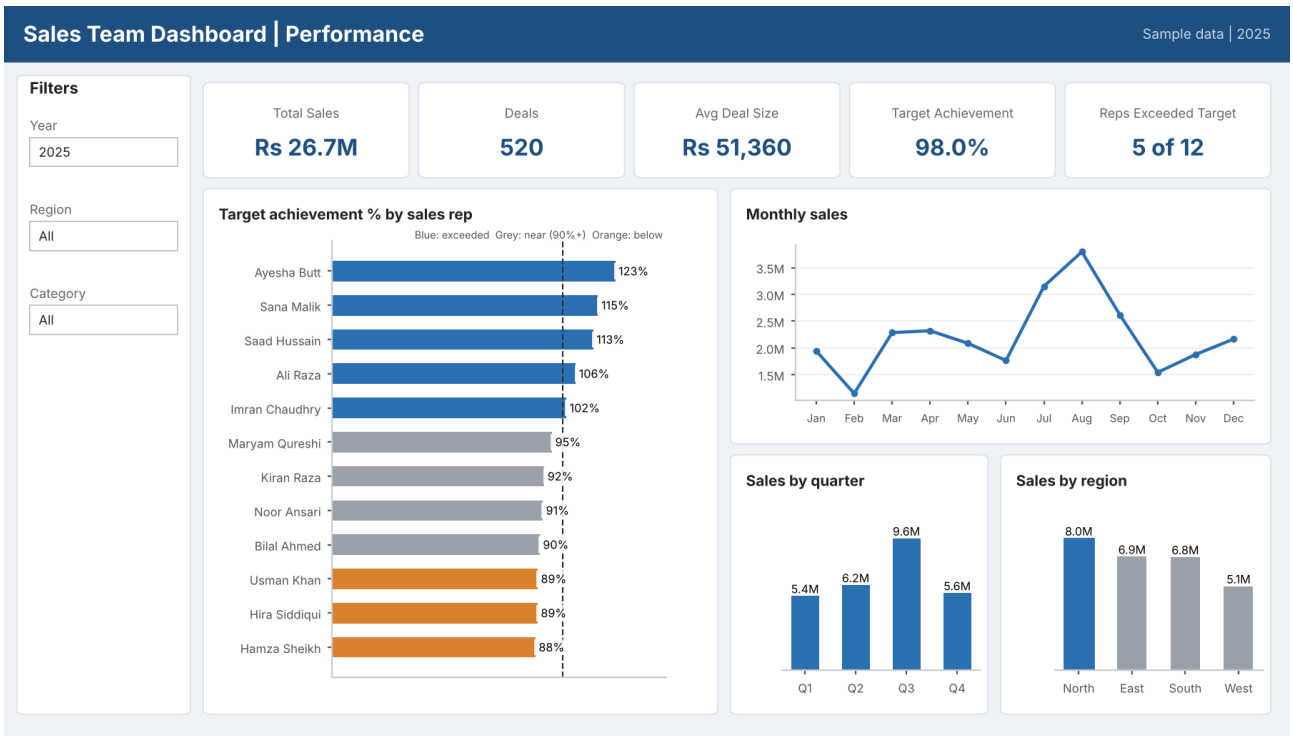
2. Measures (DAX)

Every number on the dashboard comes from a measure. The values below are the results for the full year, calculated from the same data. The DAX code is in Appendix B.

Measure	What it calculates	Value
Total Sales	Sum of sale amounts	Rs 26.71M
Deals	Number of sales rows	520
Units Sold	Total quantity	1,317
Avg Deal Size	Sales per deal	Rs 51,360
Annual Target	Sum of monthly targets x 12	Rs 27.24M
Achievement %	Sales divided by target	98.0%
Target Status	Exceeded / Near target / Below target	per rep
Reps Exceeded Target	Reps at 100% or more	5
Sales Previous Month	Sales of the month before (Dec 2025 view)	Rs 1.88M
MoM Growth %	Change vs previous month (Dec 2025 view)	+15.3%
Stock On Hand	Units in warehouse	310
Low Stock Products	Stock at or below reorder level	6
Units Sold Last 90 Days	Quantity in the last 90 days of data	333
Never Sold Products	Products with no sales	1

3. Dashboard design

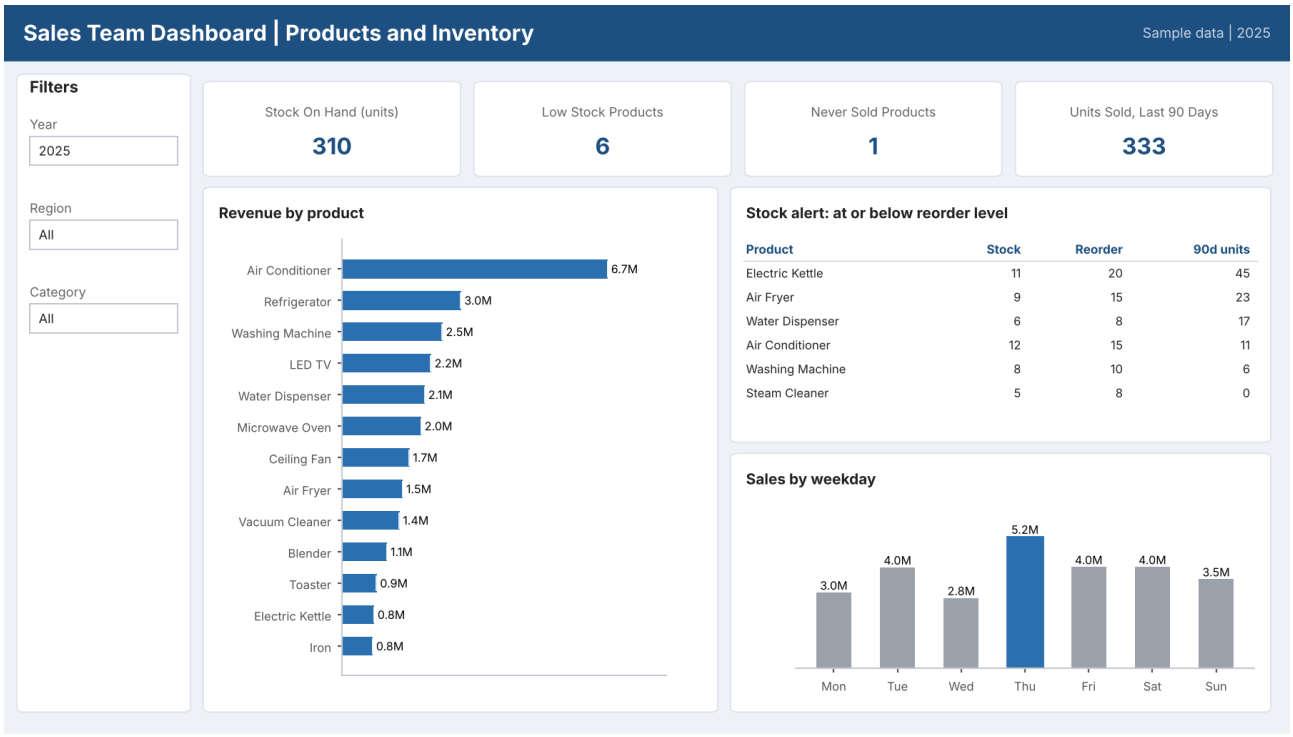
Page 1: Sales Performance



Dashboard preview (design mockup with real values from the data).

Visual	Type	Fields
Card x5	Card	Total Sales, Deals, Avg Deal Size, Achievement %, Reps Exceeded Target
Target achievement by rep	Bar chart	Axis: Employees[Name] | Values: Achievement % | Colour rule: Target Status
Monthly sales	Line chart	Axis: Calendar[Month] | Values: Total Sales
Sales by quarter	Column chart	Axis: Calendar[Quarter] | Values: Total Sales
Sales by region	Column chart	Axis: Employees[Region] | Values: Total Sales
Filters	Slicers	Calendar[Year], Employees[Region], Products[Category]

Page 2: Products and Inventory



Dashboard preview (design mockup with real values from the data).

Visual	Type	Fields
Card x4	Card	Stock On Hand, Low Stock Products, Never Sold Products, Units Sold Last 90 Days
Revenue by product	Bar chart	Axis: Products[Name] | Values: Total Sales
Stock alert	Table + visual filter	Products[Name], Stock, ReorderLevel, Units Sold Last 90 Days | Filter: Low Stock = 1
Sales by weekday	Column chart	Axis: Calendar[Weekday] | Values: Total Sales

Design guidelines

• Canvas 16:9 (1280 x 720), light grey page background, white tiles with a thin border.
• Colours: dark blue for headers, blue for the main series, grey for context, orange only to flag problems or highlights. Font: Segoe UI.
• KPI cards on top, trends in the middle, breakdowns at the bottom, filters on the left.
• Every visual has a clear title; no more than 6 to 7 visuals per page.

4. Insights and recommendations

What the data shows

• Total sales were Rs 26.71M from 520 deals, which is 98.0% of the combined annual target.

• 5 of 12 reps exceeded their target; 4 were near target and 3 were below. Best: Ayesha Butt (123%); lowest: Hamza Sheikh (88%).

• Q3 was the strongest (Rs 9.6M) and Q1 the weakest (Rs 5.4M).

• North is the top region (Rs 8.0M) and West the lowest (Rs 5.1M).

• Air Conditioner earns the most revenue (Rs 6.7M).

• 6 products are at or below their reorder level; the most urgent is Electric Kettle (stock 11, 45 units sold in the last 90 days).

• Steam Cleaner had no sales in the year.

Recommended actions

• Coach the 3 below-target reps using the top reps' approach.

• Re-order Electric Kettle first, because demand is highest among low-stock items.

• Review the never-sold product: promote it, change its price or stop stocking it.

• Add sales support in West, the lowest region.

• Plan campaigns for Q1, the slowest quarter.

5. How to build it in Power BI Desktop

Create the folder `C:\PowerBI\SalesTeam` and save these three CSV files in it:

File	Table	Columns	Rows
sales.csv	Sales	SaleID, EmpID, ProductID, SaleDate, Quantity, Amount	520
employees.csv	Employees	EmpID, Name, Region, MonthlyTarget	12
products.csv	Products	ProductID, Name, Category, Price, Stock, ReorderLevel	14

1. Home > Get data > Text/CSV. Load the three files. (Or open Transform data > New Source > Blank Query > Advanced Editor and paste the Power Query code from Appendix B.)

2. In Power Query check the column types (dates as Date, amounts as Decimal) and click Close and Apply.

3. Modeling > New table. Paste the Calendar code from Appendix B. Mark it as the date table and sort Month by Month Number.

4. Open Model view and create the relationships from the table in section 1 (drag the key column from the Sales table to the dimension).

5. Right-click the Sales table > New measure. Paste each measure from Appendix B. Set formats (currency, percentage) in the Measure tools tab.

6. Report view > View > Page view > Actual size; set Format page > Canvas settings to 16:9 and a light grey background.

7. Add the visuals from the tables in section 3. Drag the fields exactly as listed; use Top N in the Filters pane where noted.

8. Add slicers and align everything to a grid (Format > Align). Add titles, turn off unnecessary gridlines and keep colours consistent.

9. Duplicate the page for page 2, replace the visuals, then File > Save. Export with File > Export > PDF for submission.

Appendix A: Sales list

First 45 records out of 520 rows of the Sales table (with names added from the dimension tables).

Saleid	Saledate	Rep	Region	Product	Quantity	Amount
1	2025-01-01	Ayesha Butt	South	Refrigerator	2	176,400
2	2025-01-02	Ayesha Butt	South	Washing Machine	1	68,400
3	2025-01-03	Noor Ansari	South	Electric Kettle	1	4,500
4	2025-01-03	Kiran Raza	West	Ceiling Fan	1	11,875
5	2025-01-04	Hira Siddiqui	West	Refrigerator	1	98,000
6	2025-01-04	Hamza Sheikh	East	Blender	2	12,960
7	2025-01-04	Noor Ansari	South	Microwave Oven	1	25,200
8	2025-01-04	Ayesha Butt	South	Toaster	1	4,940
9	2025-01-04	Ayesha Butt	South	Iron	3	10,830
10	2025-01-05	Maryam Qureshi	West	Toaster	4	20,800
11	2025-01-06	Noor Ansari	South	Ceiling Fan	5	56,250
12	2025-01-07	Ayesha Butt	South	Refrigerator	1	98,000
13	2025-01-07	Ayesha Butt	South	Ceiling Fan	2	25,000
14	2025-01-08	Usman Khan	East	Water Dispenser	3	86,400
15	2025-01-09	Maryam Qureshi	West	Air Conditioner	2	243,000
16	2025-01-09	Kiran Raza	West	Electric Kettle	1	4,050
17	2025-01-10	Imran Chaudhry	North	Microwave Oven	1	28,000
18	2025-01-10	Usman Khan	East	Electric Kettle	5	21,375
19	2025-01-11	Noor Ansari	South	Toaster	5	26,000
20	2025-01-12	Imran Chaudhry	North	Blender	3	21,600
21	2025-01-12	Bilal Ahmed	North	Iron	4	15,200
22	2025-01-16	Saad Hussain	East	Vacuum Cleaner	3	72,000
23	2025-01-18	Maryam Qureshi	West	Iron	1	3,800
24	2025-01-20	Bilal Ahmed	North	Ceiling Fan	3	37,500
25	2025-01-21	Maryam Qureshi	West	Water Dispenser	2	60,800
26	2025-01-21	Kiran Raza	West	Washing Machine	1	72,000
27	2025-01-23	Sana Malik	South	Iron	4	14,440
28	2025-01-23	Noor Ansari	South	Electric Kettle	2	9,000
29	2025-01-23	Maryam Qureshi	West	Electric Kettle	1	4,500
30	2025-01-23	Noor Ansari	South	Ceiling Fan	3	33,750
31	2025-01-24	Kiran Raza	West	Toaster	2	9,360
32	2025-01-24	Bilal Ahmed	North	Microwave Oven	3	79,800
33	2025-01-27	Imran Chaudhry	North	Blender	4	28,800
34	2025-01-28	Sana Malik	South	LED TV	1	61,750
35	2025-01-30	Maryam Qureshi	West	Ceiling Fan	1	12,500
36	2025-01-30	Noor Ansari	South	Refrigerator	1	98,000

Saleid	Saledate	Rep	Region	Product	Quantity	Amount
37	2025-01-31	Kiran Raza	West	Toaster	3	15,600
38	2025-01-31	Sana Malik	South	Air Conditioner	2	270,000
39	2025-02-01	Ali Raza	North	Water Dispenser	2	60,800
40	2025-02-02	Hamza Sheikh	East	Ceiling Fan	4	50,000
41	2025-02-03	Bilal Ahmed	North	Iron	4	14,440
42	2025-02-03	Hira Siddiqui	West	Vacuum Cleaner	2	43,200
43	2025-02-04	Usman Khan	East	Vacuum Cleaner	1	24,000
44	2025-02-05	Kiran Raza	West	Iron	3	11,400
45	2025-02-06	Noor Ansari	South	Microwave Oven	2	53,200

Appendix B: Power Query (M) and DAX code

Copy-paste ready. Part 1 loads the CSV files in Power Query; part 2 is DAX.

```
// ===== PART 1: POWER QUERY (M) - one query per table =====

// ----- Query: Sales -----
let
    Source = Csv.Document(
        File.Contents("C:\PowerBI\SalesTeam\sales.csv"),
        [Delimiter = ",", Columns = 6, Encoding = 65001, QuoteStyle = QuoteStyle.Csv]),
    #"Promoted Headers" = Table.PromoteHeaders(Source, [PromoteAllScalars = true]),
    #"Changed Type" = Table.TransformColumnTypes(#"Promoted Headers", {
        {"SaleID", Int64.Type},
        {"EmpID", Int64.Type},
        {"ProductID", Int64.Type},
        {"SaleDate", type date},
        {"Quantity", Int64.Type},
        {"Amount", type number}
    })
in
    #"Changed Type"

// ----- Query: Employees -----
let
    Source = Csv.Document(
        File.Contents("C:\PowerBI\SalesTeam\employees.csv"),
        [Delimiter = ",", Columns = 4, Encoding = 65001, QuoteStyle = QuoteStyle.Csv]),
    #"Promoted Headers" = Table.PromoteHeaders(Source, [PromoteAllScalars = true]),
    #"Changed Type" = Table.TransformColumnTypes(#"Promoted Headers", {
        {"EmpID", Int64.Type},
        {"Name", type text},
        {"Region", type text},
        {"MonthlyTarget", Int64.Type}
    })
in
    #"Changed Type"

// ----- Query: Products -----
let
    Source = Csv.Document(
        File.Contents("C:\PowerBI\SalesTeam\products.csv"),
        [Delimiter = ",", Columns = 6, Encoding = 65001, QuoteStyle = QuoteStyle.Csv]),
    #"Promoted Headers" = Table.PromoteHeaders(Source, [PromoteAllScalars = true]),
    #"Changed Type" = Table.TransformColumnTypes(#"Promoted Headers", {
        {"ProductID", Int64.Type},
        {"Name", type text},
        {"Category", type text},
        {"Price", type number},
        {"Stock", Int64.Type},
        {"ReorderLevel", Int64.Type}
    })
in
    #"Changed Type"

// ===== PART 2: DAX =====

-- ===== Calendar table (Modeling > New table) =====
Calendar =
ADDCOLUMNS (
    CALENDAR ( DATE ( 2025, 1, 1 ), DATE ( 2025, 12, 31 ) ),
    "Year", YEAR ( [Date] ),
    "Month Number", MONTH ( [Date] ),
    "Month", FORMAT ( [Date], "MMM" ),
    "Quarter", "Q" & ROUNDUP ( MONTH ( [Date] ) / 3, 0 ),
    "Weekday", FORMAT ( [Date], "ddd" ),
    "Weekday Number", WEEKDAY ( [Date], 2 )
)

-- ===== Measures (Modeling > New measure) =====
Total Sales = SUM ( Sales[Amount] )

Deals = COUNTROWS ( Sales )

Units Sold = SUM ( Sales[Quantity] )

Avg Deal Size = DIVIDE ( [Total Sales], [Deals] )

Annual Target = SUM ( Employees[MonthlyTarget] ) * 12

Achievement % = DIVIDE ( [Total Sales], [Annual Target] )

Target Status =
SWITCH (
    TRUE (),
    [Achievement %] >= 1, "Exceeded",
```

```
    [Achievement %] >= 0.9, "Near target",
    "Below target"
)

Reps Exceeded Target =
COUNTROWS ( FILTER ( VALUES ( Employees[Name] ), [Achievement %] >= 1 ) )

Sales Previous Month =
CALCULATE ( [Total Sales], DATEADD ( 'Calendar'[Date], -1, MONTH ) )

MoM Growth % =
DIVIDE ( [Total Sales] - [Sales Previous Month], [Sales Previous Month] )

Stock On Hand = SUM ( Products[Stock] )

Low Stock Products =
COUNTROWS ( FILTER ( Products, Products[Stock] <= Products[ReorderLevel] ) )

Units Sold Last 90 Days =
VAR LastDate = CALCULATE ( MAX ( Sales[SaleDate] ), ALL ( Sales ) )
RETURN
    CALCULATE (
        SUM ( Sales[Quantity] ),
        DATESINPERIOD ( 'Calendar'[Date], LastDate, -90, DAY )
    )

Never Sold Products =
COUNTROWS ( FILTER ( Products, ISBLANK ( CALCULATE ( COUNTROWS ( Sales ) ) ) ) )
```